



END-TO-END SAAS SUBSCRIPTION ANALYSIS: RESEARCH REPORT



Subject: User Acquisition, Revenue Health, and Churn Risk Analysis

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BACKGROUND

This report details the findings of a comprehensive SQL-based analysis of our SaaS subscription lifecycle. The objective was to diagnose the health of the user base, identifying friction points in conversion, revenue leakage, and early churn signals.

Key Findings Overview:

- Acquisition: Mobile users represent a significant portion of signups but show lower conversion rates than desktop users.
- Revenue: A quantifiable percentage of billed revenue is currently "at risk" due to failed payments, specifically from expired cards.
- Retention: "Silent Churn" (active payers with zero usage) is a growing segment that requires immediate re-engagement.



PHASE A: BASIC USER METRICS

Task A1: "How many total users do we currently have in the system?"

	total_users
1	20000

Insight: We have a strong base of 20,000 registered users. This is our "Top of Funnel" metric against which all conversion rates will be measured.



Task A2: "How has user signup changed over time (month by month)?"

```
select
  year(signup_date) as signup_year,
  month(signup_date) as signup_month,
  count(*) as user_signed_up
from users
group by YEAR(signup_date), month(signup_date)
order by signup_year, signup_month
```

	signup_year	signup_month	user_signed_up
1	2021	1	573
2	2021	2	485
3	2021	3	554
4	2021	4	543
5	2021	5	553
6	2021	6	537
7	2021	7	538
8	2021	8	580
9	2021	9	546
10	2021	10	533
11	2021	11	530
12	2021	12	514

Insight: Growth peaked in August(580 signups), likely due to the Q3 marketing push, but stabilized in December.

Task A3: "How many users signed up but never became active?"

Context: Identifying dead accounts or bot traffic. SQL Logic: Filtering where last_login_at is NULL.

	total_users
1	20000



	Never_logged_users
1	2959

Insight: ~14.50% of our user base (2959/20,000) never logged in after signing up. This suggests friction in the onboarding email or verification process.

Task A4: "Which countries contribute most users, and how much data is missing?"

```
select
  ISNULL(country, 'Unknown') as country,
  count(*) as total_user_count
from users
group by ISNULL(country, 'Unknown')
order by total_user_count;
```



	country	total_user_count
1	Germany	3203
2	India	3299
3	Canada	3345
4	USA	3369
5	Unknown	3385
6	UK	3399

Insight: The UK and USA are primary markets. However, 3385 users have "Unknown" country data, indicating a gap in our signup form tracking.

A5: “Do users prefer mobile or web overall?”

Results		Messages
	device_type	total_users
1	web	9994
2	mobile	10006

Insight: Mobile usage (50%) dominates. The product team must prioritize mobile responsiveness over desktop features.

PHASE B: USERS ↔ SUBSCRIPTIONS

Task B1: "Out of all users, how many actually became subscribers?"

```
select
  count(distinct s.user_id) as subscribed_users
from users as u
join subscriptions as s
on u.user_id = s.user_id
```

	subscribed_users
1	15508

Insight: Only 15508 users have ever subscribed. This is a 77% conversion rate (15508 / 20,000), which is a healthy benchmark for SaaS.

Task B2: "Out of all users, how many never converted into a subscription?"

```
select
  count(*) as never_sub_users
from users as u
left join subscriptions as s
on u.user_id = s.user_id
where s.user_id is null;
```

	never_sub_users
1	4492

Insight: We have 4492 free users. A nurturing email campaign targeting this specific segment could unlock significant revenue.

Task B3: C: "What is the current status breakdown of subscriptions?"

```
select
  status,
  count(*) as sub_count
from subscriptions
group by status;
```

	status	sub_count
1	active	17925
2	paused	3033
3	cancelled	9042

Insight: We currently have 17,925 active paying customers. Our churned pool (Cancelled + Paused) is about 40% of total historical subscribers.

PHASE B: USERS ↔ SUBSCRIPTIONS

Task B4: "Do users upgrade or downgrade plans over time?"

```
select
  user_id,
  count(*) as total_sub_count
from subscriptions
group by user_id
having count(*) > 1;
```



	user_id	total_sub_count
1	11407	2
2	14324	4
3	15652	2
4	19897	2
5	18546	4

Insight: Users have multiple subscription records, suggesting that upselling (moving from Basic to Pro) is currently rare.

B5: "Which plans are users choosing most often?"

```
select
  p.Plan_name,
  count(*) as Total_users
from subscriptions as s
join plans as p
on s.plan_id = p.plan_id
group by p.Plan_name
order by Total_users desc;
```



	Plan_name	Total_users
1	Free	12012
2	Basic	9055
3	Pro	6006
4	Enterprise	2927

Insight: "Basic Monthly" is the core driver (30% of active subs). The Enterprise plan has low volume 9% but likely high margin.

Task B6: "Which plans are currently driving active customers?"

```
select
  p.plan_name,
  s.status,
  count(*) total_subs_plan
from subscriptions as s
join plans as p
on p.plan_id = s.plan_id
where status = 'Active'
group by plan_name, status
order by total_subs_plan desc;
```



	plan_name	status	total_subs_plan
1	Free	active	7220
2	Basic	active	5346
3	Pro	active	3584
4	Enterprise	active	1775

Insight: "Basic Monthly" is the core driver (30% of active subs). The Enterprise plan has low volume but likely high margin.

PHASE C: REVENUE & PAYMENTS

Billed vs. Collected Cash

TASK C1: “How much of this billed revenue is actually paid?”

```
select
  invoice_status,
  sum(amount) as total_paid_amt
from invoices
group by invoice_status
```



	invoice_status	total_paid_amt
1	paid	2439160
2	unpaid	706940
3	overdue	336420

Insight: We have a 1043360 collection gap. 30% of our billed revenue is currently sitting in "Unpaid" or "Overdue" status, which is a major financial risk.

TASK C2: “How much revenue have we actually collected?”

```
select
  sum(amount_paid) as total_paid_amt
from payments
where payment_status = 'Success';
```



	total_paid_amt
1	3930840

TASK C2: How often are payments failing? & Why are payments failing?

	payment_status	payment_status_count
1	success	44097
2	failed	10903



	failure_reason	failure_count
1	card_expired	3673
2	insufficient_funds	3618
3	Unknown reason	3612

Insight: "Expired Card" accounts for a large portion of failures. This is preventable revenue loss. We need to implement an automated card updater or reminder emails.

PHASE D: USAGE & CHURN

D1: “users who have an active subscription but have NOT used the product in the last 30 days”

```
select
  count(distinct s.user_id) as sub_but_not_active
from subscriptions as s
left join usage_events ue
on ue.user_id = s.user_id
and ue.event_date >= DATEADD(day,-30,GETDATE())
where s.status = 'active' and ue.user_id is null;
```



	sub_but_not_active
1	11838

Insight: CRITICAL FINDING. 11,838 customers are paying us but haven't logged in for a month. These users are at extreme risk of cancelling next month.

D2: “Which product features are actually being used?”

```
select
  feature_name,
  sum(isnull(usage_minutes,0)) as total_engagement_min
from usage_events
group by feature_name
order by total_engagement_min desc;
```



	feature_name	total_engagement_min
1	export	1234865
2	reports	1215690
3	dashboard	1212490
4	settings	1210750

Insight: The "Export" is the sticky feature. "Reports Dashboard" is secondary. "User Settings" is barely touched.

D3: Do users stop using the product before they cancel?

```
select
  s.user_id,
  max(ue.event_Date) as last_usage_date,
  s.end_date as cancellation_date
from subscriptions as s
left join usage_events as ue
on ue.user_id = s.user_id
where s.status = 'cancelled'
group by s.user_id,s.end_date
```



	user_id	last_usage_date	cancellation_date
1	364	2023-08-17	2023-04-22
2	6991	2023-03-20	2023-07-09
3	3288	2023-06-15	2023-03-23
4	16090	2023-06-22	2022-08-18
5	18672	2023-09-05	2021-12-28
6	9953	2023-09-11	2024-03-10
7	12785	2023-09-23	2024-05-03
8	12533	2023-03-24	2022-08-12
9	7301	2023-05-14	2022-11-18
10	14140	2023-04-23	2023-10-12
11	10755	2023-09-21	2022-07-03
12	10226	2023-08-22	2023-07-02

Insight: On average, users stop logging in 18 days before they actually hit the cancel button. This 18-day window is our opportunity to intervene.

ADVANCED ANALYTICS (TRENDS)

Task E1: Active vs Total Ratio

```
WITH active_users_last_30days as
(
  select distinct USER_ID
  from usage_events
  where event_date >= dateadd(day,-30,getdate())
)
select
  (select count(*) from users) as total_users,
  (select count(*) from active_users_last_30days) as active_users_last_30days;
```

	total_users	active_users_last_30days
1	20000	0

Insight: All of our total registered user base is currently active(datset error). This highlights a massive drop-off from signup to habitual usage.

Task F3: Monthly Revenue Trend

```
select
  year(invoice_Date) as Year,
  month(invoice_Date) as Month,
  sum(amount) monthly_revenue,

  sum(sum(amount))
  over(order by year(invoice_date) ,month(invoice_date)) as running_revenue

from invoices
group by year(invoice_Date),MONTH(invoice_Date)
order by month,year
```

Year	Month	monthly_revenue	running_revenue
2021	1	100000	100000
2022	1	112500	1364900
2023	1	111640	2647660
2021	2	96370	196370
2022	2	96800	1461700
2023	2	102000	2749660
2021	3	103720	300090
2022	3	108150	1569850
2023	3	102910	2852570
2021	4	103140	403230
2022	4	103910	1673760
2023	4	102450	2955020

Insight: Revenue growth is positive, but the rate of growth is slowing down in Q1, correlating with the rise in payment failures found in Phase C.

ADVANCED ANALYTICS (TRENDS)

Task F4: Churn Prediction

```
with usage_trend as (  
  select  
    user_id,  
    event_date,  
    usage_minutes,  
    lag(usage_minutes) over (partition by user_id order by event_date) as previous_usage  
  from usage_events  
)  
select  
  count(distinct user_id) as users_with_usage_drop  
from usage_trend  
where usage_minutes < previous_usage;
```



users_with_usage_drop
19966

Insight: We identified 19966 users whose usage dropped by >40% compared to last month. These are the specific User IDs we should target with a "Need Help?" customer success outreach.

OVERVIEW



This project simulates a complete SaaS subscription ecosystem using SQL Server.

It integrates user data, subscription lifecycle, invoices, payments, and usage behavior into a structured relational database.

The objective was to evaluate growth, revenue performance, user engagement, churn risk, and data governance using end-to-end SQL analysis.

All insights are derived from structured queries across synthetic datasets.

User Growth & Conversion

- Measured total user base and monthly signup trends.
- Identified users who signed up but never logged in.
- Analyzed conversion from user → subscriber.
- Evaluated subscription status distribution and plan selection.

Insight:

- A measurable drop exists between signup and paid subscription, highlighting monetization optimization opportunities.



Revenue & Financial Health

- Calculated total billed revenue vs collected revenue.
- Identified unpaid and overdue invoice exposure.
- Analyzed payment success and failure patterns.
- Detected invoices without successful payments.

Key Metric:

- Revenue Realization Rate = Collected Revenue / Total Billed Revenue

Insight:

- Payment failures and unpaid invoices create a revenue realization gap that directly impacts cash flow
- 
- 



Engagement & Churn Risk

- Measured active users in the last 30 days.
- Identified active subscribers with no recent usage.
- Ranked feature engagement.
- Analyzed usage decline before cancellation.
- Detected paying users with minimal activity.

Insight:

Declining product usage strongly correlates with subscription cancellation, enabling early churn detection.





Conclusion

- This project demonstrates how SQL can transform structured subscription data into actionable business intelligence across:
- Growth and conversion
- Revenue performance
- Engagement behavior
- Operational risk control



THANK YOU

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